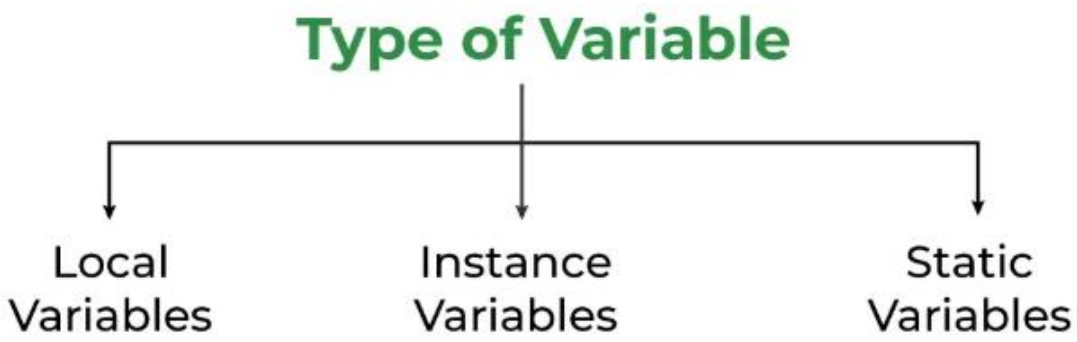
Variable:

- A variable is a name assigned to a value that is stored inside the system memory. The value can be updated during the program execution.



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1. Local Variables

A variable defined within a block or method or constructor is called a local variable.

- **The Local variable is created at the time of declaration and destroyed after exiting from the block or when the call returns from the function.**
- **The scope of these variables exists only within the block in which the variables are declared, i.e., we can access these variables only within that block.**
- **Initialization of the local variable is mandatory before using it in the defined scope.**

Example 1

```
class Test {

  public static void main(String[] args) {
```

```
    int x = 10;//local variable

    System.out.println(x);

  }
}
```

Output

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
10
PS C:\Users\Admin\Desktop\JQ Ajax>
```

Example 2:

```
class Test {

  public static void main(String[] args) {
    int x;//local variable

    System.out.println(x);

  }
}
```

In the above example we will get compile time error.

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
Test.java:7: error: variable x might not have been initialized
    System.out.println(x);
                       ^
```

2. Instance Variables

Instance variables are non-static variables and are declared in a class outside of any method, constructor, or block.

- **As instance variables are declared in a class, these variables are created when an object of the class is created and destroyed when the object is destroyed.**
- **Unlike local variables, we may use access specifiers for instance variables. If we do not specify any access specifier, then the default access specifier will be used.**

Example:

```java
class Test {
  int x;// instance variable

  public static void main(String[] args) {
    Test t = new Test();
    t.x = 10;
    System.out.println(t.x);

  }
}
```

Output:

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
10
```

Example 2:

```java
class Test {
  int x;// instance variable

  public static void main(String[] args) {
    Test t = new Test();

    System.out.println(t.x);

  }
}
```

Output:

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
0
```

Default Value of Instance Variable:

Instance variable type	Default values
boolean	false
byte	(byte) 0
short	(short) 0
int	0
double	0.0d
float	0.0
long	0L
Object	null
char	\u0000

3. Static Variables

Static variables are also known as class variables.

- **These variables are declared similarly to instance variables. The difference is that static variables are declared using the static keyword within a class outside of any method, constructor, or block.**
- **Unlike instance variables, we can only have one copy of a static variable per class, irrespective of how many objects we create.**
- **Static variables are created at the start of program execution and destroyed automatically when execution ends.**

Example:

```java
class Test {
  static int x;// static variable

  public static void main(String[] args) {

    System.out.println(x);

  }
}
```

Output

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
0
```

Example :

```java
class Test {
  static int x = 10;// static variable

  public static void main(String[] args) {

    System.out.println(Test.x);
    System.out.println(x);

  }
}
```

Output

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
10
10
```

What is Primitive Variable:

In Java, a primitive variable is a variable that holds a fundamental data type directly, rather than a reference to an object.

Direct Value Storage:
Primitive variables store the actual value of the data type, unlike reference variables that store the memory address of an object

For Example:

```java
int age = 25;
double salary = 50000.50;
char grade = 'A';
boolean isStudent = true;
```

Reference Variable:

In Java, a reference variable is a variable that stores the memory address of an object, rather than the object itself. It acts like a pointer, allowing you to access and manipulate the object's data and behavior.

- **Object Access:**

  Reference variables allow you to interact with objects created from classes.
- **Memory Management:**

  They help manage memory efficiently by storing only the address, not the entire object.

- **Reference variables are fundamental to working with objects in Java.**
- **They allow you to access and manipulate objects without duplicating the object's data in memory.**

Exmple:

```
class Test {
  int x;

  public static void main(String[] args) {

    Test t1 = new Test();
    t1.x = 20;

    Test tr = t1;
    System.out.println(t1.x);
    System.out.println(tr.x);

  }
}
```

Output

```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
20
20
```

Example 2:

```
class Test {
  int x;

  public static void main(String[] args) {
```

```java
    Test t1 = new Test();
    t1.x = 20;

    Test tr = t1;
    System.out.println(t1.x);
    System.out.println(tr.x);

    tr.x = 100;
    System.out.println(t1.x);
    System.out.println(tr.x);
  }
}
```

Output

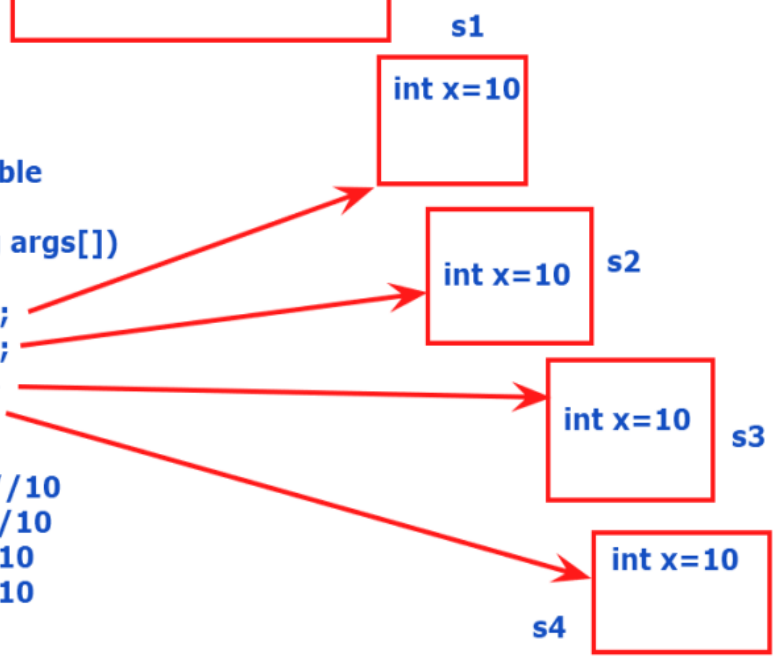
```
PS C:\Users\Admin\Desktop\JQ Ajax> javac Test.java
PS C:\Users\Admin\Desktop\JQ Ajax> java Test
20
20
100
100
```

Instance Variable

```
class Student
{
    int x=10;// Instance variable

 public static void main(String args[])
{
    Student s1=new Student();
    Student s2=new Student();
   Student s3=new Student();
  Student s4=new Student();

    System.out.println(s1.x); //10
   System.out.println(s2.x); //10
  System.out.println(s3.x);//10
  System.out.println(s4.x);//10
}
}
```



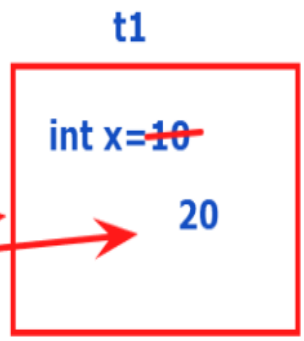
Instance Variable Example 2

```
class Test {
 int x = 10;

 public static void main(String[] args) {

  Test t1 = new Test();
  t1.x = 20;
  System.out.println(t1.x);   //20

 }
}
```

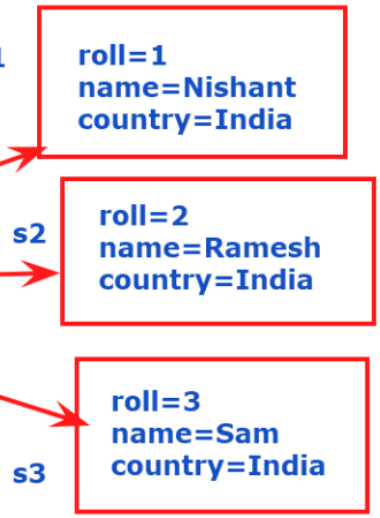


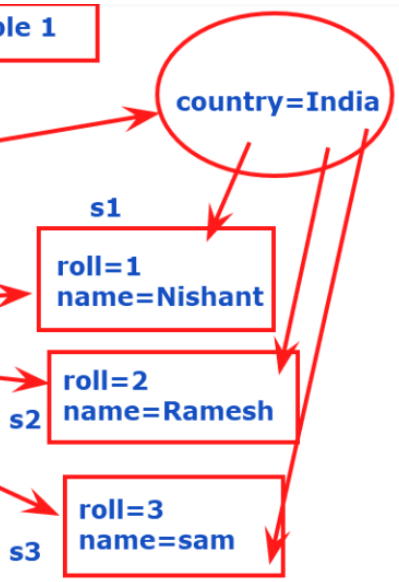
Instance Variable Example 3

```
public class Student {
  int roll;
  String name;
  String country;

  public static void main(String[] args) {

    Student s1 = new Student();
    Student s2 = new Student();
    Student s3 = new Student();
  }
}
```



Static Variable example 1

```
public class Student {
  int roll;
  String name;
  static String country;

  public static void main(String[] args) {

    Student s1 = new Student();
    Student s2 = new Student();
    Student s3 = new Student();
  }
}
```